

The Goose *SDK.*

BUILD ON THE AGENT, NOT AROUND IT

Goose is a single compiled binary — agent runtime, model access, tool execution, extensions, sessions — all included.

The SDK lets you embed it, connect to it, and extend it. Build your experience; let Goose handle the hard parts.

Months of plumbing, *or* a single import.

Most teams spend a quarter wiring up infrastructure before they ship a single agent feature. The SDK shortens that to an afternoon.

WITHOUT IT

The long road

- Wire up LLM providers — auth, streaming, retries, fallbacks
- Build session management, context windows, compaction
- Implement tool execution, sandboxing, permissions, security
- Write your own extension and plugin system
- Handle model switching, cost tracking, cancellation

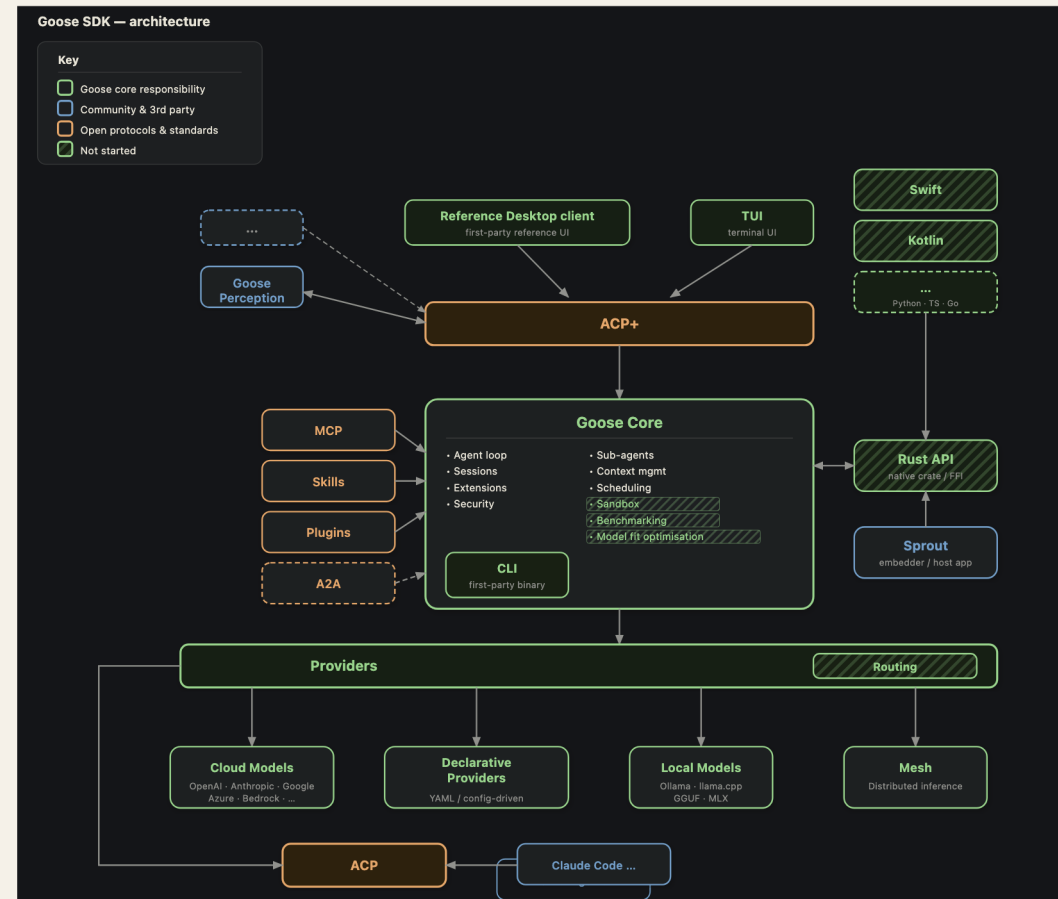
WITH THE SDK

The short one

- *All providers* — Anthropic, OpenAI, Bedrock, Databricks, xAI, Groq, Ollama, llama.cpp...
- *Security* — prompt injection detection, tool permissions, malware checks
- *Context management* — auto-compaction, token counting, cost tracking
- *Extensions* — MCP + rMCP, Skills, Recipes, lifecycle hooks
- *Sub-agents, scheduling, sandboxed execution*

One compiled binary. `brew`, `npm`, `apt`, `rpm`, Docker, or a single download.

What is *inside* the binary.



— Frontends connect via ACP+. A Rust API and language bindings open the kernel of the agent; Sprout shows what a third-party app looks like on top. Striped boxes are in flight. —

The motion of the machine.



— Inference, tool call, execute, respond — repeat. The application layer intercepts at every stage. —

How it *works*.

I. DISTRIBUTION

A single binary

Compiled Rust. Acquire it however you prefer:

- `brew install block-goose-cli`
- `npm install @aaif/goose-sdk`
- `apt` / `rpm` / `.deb` / `.flatpak`
- Docker, MSI, NSIS, curl script
- Built-in `goose update`

II. TRANSPORT

One protocol, three pipes

- *Stdio* — spawn `goose acp`, pipe JSON
- *WebSocket* — connect to a running server
- *HTTP + SSE* — for web services

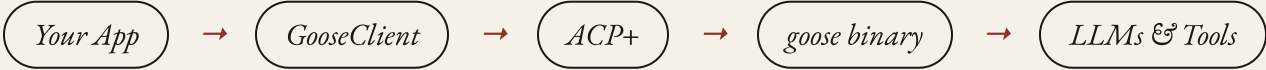
Daemon, child process, or CLI tool — your app connects and sends prompts.

III. PROTOCOL — ACP+

JSON-RPC, refined

The "+" denotes Goose's extensions to the open ACP standard:

- Provider & model management
- Extension hot-loading
- Config & secrets
- Session export & import



A full agent, in *twenty lines*.

```
import { spawn } from "node:child_process";
import { Readable, Writable } from "node:stream";
import { ndJsonStream, PROTOCOL_VERSION } from "@agentclientprotocol/sdk";
import { GooseClient } from "@aaif/goose-sdk";
import { resolveGooseBinary } from "@aaif/goose-sdk/node";

// 1. Spawn the goose binary
const proc = spawn(resolveGooseBinary(), ["acp"], { stdio: ["pipe", "pipe", "ignore"] });
const stream = ndJsonStream(Writable.toWeb(proc.stdin!), Readable.toWeb(proc.stdout!));

// 2. Create the client
const client = new GooseClient(() => ({
  sessionUpdate: async (p) => {
    if (p.update.sessionUpdate === "agent_message_chunk" && p.update.content.type === "text")
      process.stdout.write(p.update.content.text); // stream text to your UI
  },
  requestPermission: async (p) => ({
    outcome: { outcome: "selected", optionId: p.options[0].optionId }
  }),
}), stream);

// 3. Initialize, create session, prompt
await client.initialize({ protocolVersion: PROTOCOL_VERSION, clientInfo: { name: "my-app", version: "1.0" }, clientCapabilities: {} });
```

A full agent — tool use, streaming, sessions, any LLM provider.

Three clients, *three levels*.

Pick the altitude that fits your app. All three sit on the same Goose Core — same agent loop, same providers, same security model. The `GooseClient` + `@aaif/goose-sdk` covers the protocol surface; the Rust API covers in-process.

I. PROTOCOL · № TERMINAL

Text TUI — ACP over stdio

React Ink. Spawns `goose acp` as a child process. Any language, any runtime; process isolation, swap versions.

- Interactive TUI & one-shot mode
- Provider onboarding flow
- ~30 lines for a minimal client
- Same pattern Claude Code & friends use

II. IN-PROCESS · № DESKTOP

Reference Desktop — Rust API · FFI

Tauri + *React*. Embeds the kernel directly. Rust is the source of truth; **Swift**, **Kotlin**, and other bindings call into it via FFI. No daemon, no socket, lowest latency.

- Singleton connection with auto-reconnect
- Zustand stores, replay & live streaming
- Same crate the CLI ships with
- Mobile & desktop SDKs ride on this

III. APPS · № HEADLESS

goose_perception — pick either

Node.js daemon. Watches screen → updates wiki. Build an experience on top of ACP *or* in-process; Sprout is the first third-party Rust-API embedder.

- Multi-model: fast OCR + smart synthesis
- Sends images via ContentBlock
- Fresh sessions per task batch
- Ship your own UX, keep the engine

*New in this evolution: **Rust API**, **language bindings**, and **Sprout** as the first third-party crate consumer.*

A complete *agent runtime*, included.

SECURITY

Defence in depth

- *Prompt-injection detection* — pattern + ML
- *Tool permissions* — Auto, Approve, SmartApprove, Chat
- *Malware checks* — OSV before installing
- *Egress / exfiltration detection*
- *Extension sandboxing*

EXTENSIONS & SKILLS

A garden, not a fortress

- *MCP + rMCP* — local & remote tool servers, hot-loadable
- *Skills* — discoverable `SKILL.md` files, project & global

CONTEXT & SESSIONS

Memory, kept

- *Auto-compaction* at 80% window
- *Token counting* — tiktoken BPE
- *Cost tracking* — per-provider, cache-aware
- *Session persistence* — create, resume, fork, export
- *Chat recall* across sessions

AGENT CAPABILITIES

Beyond a single loop

- *Sub-agents* — parallel, isolated context
- *Scheduling* — background & recurring tasks
- *Sandboxed code execution* — Docker (*not started*)
- *Tool-loop detection*
- *Retry with backoff*
- *Benchmarking* — built-in eval harness (*not started*)
- *Model fit optimisation* — pick the right model per task (*not started*)

MODEL ACCESS

Every model, one surface

- *Cloud* — Anthropic, OpenAI, Bedrock, Databricks, xAI, Groq...
- *Local* — Ollama, llama.cpp, Candle

Extension *points.*

I. MCP + rMCP

Tools, on demand

Add tools at runtime via local or remote MCP servers. Goose manages the lifecycle — hot-loadable, no restart.

```
await client.extMethod(
  "_goose/extensions/add", {
    sessionId,
    name: "my-tool-server",
    uri: "http://localhost:8080",
  }
);
```

III. SKILLS

Procedural knowledge

Discoverable `SKILL.md` files with frontmatter. Project-scoped (`.goose/skills/`) or global (`~/.agents/skills/`). Loaded by the agent on demand.

II. LIFECYCLE HOOKS (*WIP*)

Policy, declared

Shell commands or HTTP calls at key lifecycle points — configured in YAML, no code changes.

- *before_tool_call* — block dangerous actions
- *after_tool_call* — audit, log, react
- *post_compact* — re-inject context
- *on_session_start / end* — setup & teardown

IV. RECIPES & SUB-AGENTS

Configurable agency

Recipes: declarative YAML — prompt, extensions, model, parameters. Shareable, deeplinkable, schedulable.

Sub-agents: child agents with isolated contexts for parallel workstreams.

A participant, *not a walled garden.*

Goose does not replace your existing stack — it consumes it. Any agent that speaks ACP becomes a tool Goose can orchestrate.



CONSUME

Bring your tools along

- *Claude Code* — available as a provider via ACP
- *Any ACP-compatible agent* — Goose can talk to it
- *MCP servers* — any language, any framework
- *rMCP* — remote tool servers over the network

BE CONSUMED

Yours to embed

- *Goose itself speaks ACP* — any ACP client can drive it
- *Embeddable* — stdio, WebSocket, or HTTP
- *CLI-friendly* — pipe in, pipe out, script it
- *Open protocol* — no vendor lock-in

The *LSP* *playbook* — one protocol, many clients.

ACP is already open. Goose's extensions (`_goose/*`) follow a deliberate path to standardisation.



TODAY — SHARED INFRASTRUCTURE

- `@agentclientprotocol/sdk` — TypeScript SDK (npm)
- `sacp` — Rust crate (symposium-acp)
- `agent-client-protocol-schema` — shared types (crates.io)
- JSON Schema definitions for every request & response

NEXT — TO BE STANDARDISED

- Provider discovery & model listing
- Extension management — add, remove, toggle
- Config & secret management
- Session export & import
- Lifecycle hooks

Battle-tested across 3+ clients before any proposal.

Build for `GooseClient` today → works with any ACP agent tomorrow.

Pattern: a *desktop* client.

```
// Singleton connection with auto-reconnect
let client: GooseClient | null = null;

export async function getClient() {
  if (client) return client;

  const wsUrl = await invoke("get_goose_serve_url");
  const stream = createWebSocketStream(wsUrl);

  client = new GooseClient(() => ({
    sessionUpdate: async (n) => {
      notificationHandler.handle(n);
    },
    requestPermission: async (p) => ({
      outcome: { outcome: "selected",
        optionId: p.options[0].optionId },
    })),
  })), stream);

  await client.initialize({ /* ... */ });
  client.closed.then(() => { client = null; });
  return client;
}

export const prompt = (sid, blocks) =>
  getClient().then(c =>
```

WHAT THIS GETS YOU

- *Lazy connection* — connects on first use
- *Auto-reconnect* — clears on close, next call reconnects
- *State in Zustand* — notifications route to stores
- *Session replay* — `loadSession()` replays history

NOTIFICATION → UI

```
switch (update.sessionUpdate) {
  case "agent_message_chunk":
    chatStore.append(sid, update.content); break;
  case "tool_call":
    chatStore.addTool(sid, update); break;
  case "usage_update":
    chatStore.updateUsage(sid, update); break;
}
```

Pattern: a *headless* service.

```
// Multi-model: cheap for OCR, expensive for synthesis
async function createSession(kind: "fast"|"smart") {
  const provider = kind === "fast"
    ? config.fastProvider : config.smartProvider;
  const model = kind === "fast"
    ? config.fastModel : config.smartModel;

  const s = await client.newSession({
    cwd: wikiDir, mcpServers: [],
    _meta: { provider },
  });

  if (model) await client.goose
    .GooseSessionProviderUpdate({
      sessionId: s.sessionId, provider, model,
    });
  return s.sessionId;
}

// Send screenshot for transcription
const sid = await createSession("fast");
await client.prompt({ sessionId: sid, prompt: [
  { type: "text", text: "Transcribe this screen" },
  { type: "image", data: screenshot.base64,
    mimeType: "image/png" },
] });
```

KEY PATTERNS

- *Session-per-task* — fresh batch, no context buildup
- *Multi-model* — different sessions, different providers
- *Stream buffer* — accumulate chunks, read on complete
- *Images as ContentBlock* — base64 alongside text

QUERY PROVIDERS AT RUNTIME

```
const providers = await client.goose
  .GooseProvidersDetails({});

providers.providers
  .filter(p => p.isConfigured)
  .forEach(p => console.log(p.name));

const models = await client.goose
  .GooseProvidersModels({
    providerName: "anthropic"
  });
```

Build on the agent, *not around it.*

Three clients prove it works. Yours could be the fourth.

I. ONE BINARY, RUNS ANYWHERE

Compiled Rust. Brew, npm, apt, rpm, Docker, Windows MSI, curl. Daemon, child process, or CLI.

II. SECURITY, BUILT IN

Prompt-injection detection, tool-permission modes, malware checks, egress monitoring, sandboxed execution.

III. EXTEND, DON'T FORK

MCP + rMCP for tools. Skills for knowledge. Recipes for config. Hooks for policy. Sub-agents for parallelism.

IV. OPEN PROTOCOL

ACP is the standard. Goose's ACP+ extensions prove out in production, then get standardised.

Reference Desktop · *desktop*

Text TUI · *terminal*

goose_perception · *headless*

`brew install block-goose-cli` — or `npm`, `apt`, `docker`, `curl`...